SPARKY'S SCIENCE EXPERIMENTS

Beans in a Jar: Watch Roots Grow Down

Life Science (plants) · Ages 5+ · 10 minutes, then watch for a week

Grow dry beans in a clear jar with cotton wool or paper towel, so you can see the roots grow down and the shoot grow up. Yes, beans grow well in cotton wool!

Safety first

Sparky always says: Don't eat the raw beans - some kinds make you sick!

- Wash your hands after touching the seeds. Throw it away if fuzzy mould grows.

You need

- A clear jar or plastic cup
- Cotton wool or paper towel
- 3-4 dry beans (sugar beans work; mielie kernels too)
- Water and a ruler

Steps

1. Fill the jar loosely with cotton wool or crumpled paper towel.
2. Make it damp all through, but pour out any water lying at the bottom.
3. Push the beans between the cotton and the glass, halfway up, so you can see them.
4. Keep the jar in a warm, bright place, out of hot sun all day. Add a little water daily.
5. Look and measure every day. In 2 to 5 days a root comes out; a green shoot follows.
6. When the roots are 2 cm long, lay the jar on its side. In a day or two the root bends down again!

If it doesn't work, try these in order

1. No root after 5 days? Keep the cotton damp (not soaking) and the jar warm - cold slows it down.
2. Mould? It is too wet: start again with fresh cotton and less water.
3. Still nothing? Old beans may not sprout - try beans from a new packet.

Why it works

Inside a seed is a tiny baby plant with its food. Water and warmth wake it up: this is germination. Roots grow down and shoots grow up because the plant can feel gravity (gravitropism). Tiny grains of starch in the root tip sink to the bottom of the cells and show the root which way is down.

Did you know? Farmers in most of South Africa plant mielies and beans when the summer rains start, because seeds need water to germinate.

Ask Sparky: Will the roots still grow down in the dark? Which grows faster, a bean or a mielie?